Evidence Analysis

1. Data Collection using Velociraptor:-

The evidence collection process was performed using **Velociraptor**, a Digital Forensics and Incident Response (DFIR) tool used for endpoint monitoring and investigation.

Due to system compatibility limitations (Windows environment), live execution screenshots could not be captured. However, the methodology followed is described below.

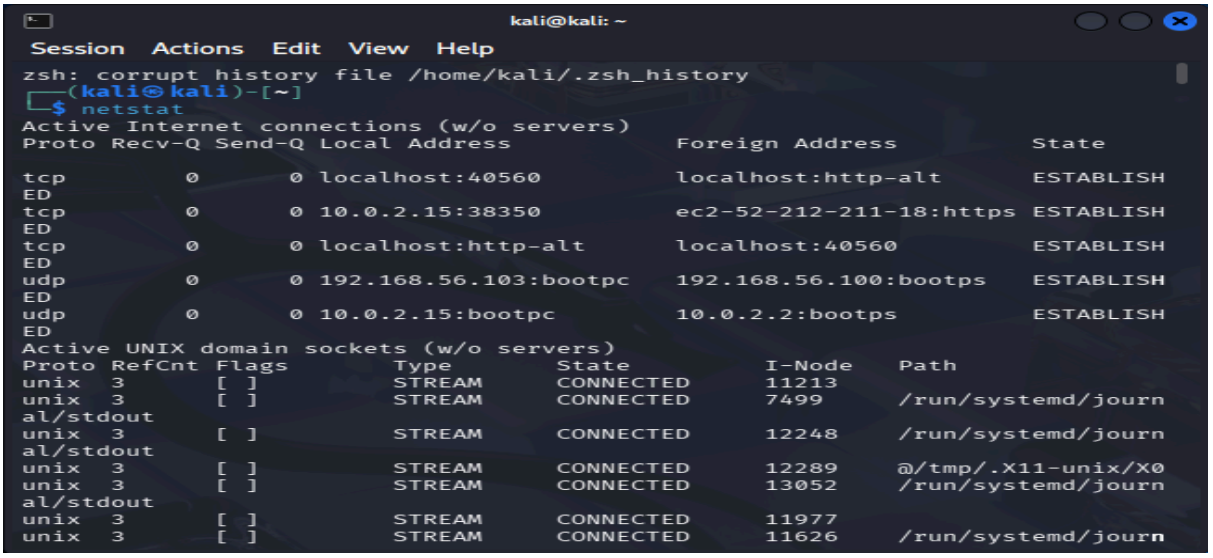
A query was executed to extract active network connections:-

- **SELECT * FROM netstat**

This query is used to collect system-level network data, including:

- Active network connections
- Source and destination IP addresses
- Port numbers
- Communication protocols

The collected data helps in identifying suspicious or unauthorized network activities within the system.



```
kali@kali: ~
Session  Actions  Edit  View  Help
zsh: corrupt history file /home/kali/.zsh_history
┌──(kali㉿kali)-[~]
└─$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 localhost:40560         localhost:http-alt      ESTABLISH
ED
tcp        0      0 10.0.2.15:38350         ec2-52-212-211-18:https ESTABLISH
ED
tcp        0      0 localhost:http-alt      localhost:40560         ESTABLISH
ED
udp        0      0 192.168.56.103:bootpc   192.168.56.100:bootps   ESTABLISH
ED
udp        0      0 10.0.2.15:bootpc        10.0.2.2:bootps         ESTABLISH
ED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  3      [ ]         STREAM     CONNECTED     11213
unix  3      [ ]         STREAM     CONNECTED     7499     /run/systemd/journ
al/stdout
unix  3      [ ]         STREAM     CONNECTED     12248    /run/systemd/journ
al/stdout
unix  3      [ ]         STREAM     CONNECTED     12289    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     13052    /run/systemd/journ
al/stdout
unix  3      [ ]         STREAM     CONNECTED     11977
unix  3      [ ]         STREAM     CONNECTED     11626    /run/systemd/journ
```

Figure 1: Sample netstat output showing active network connections

2. Identification of Suspicious Connections:-

The collected data was analyzed to identify anomalies in network activity.

The following indicators were used to detect suspicious behavior:

- Connections to **unknown IP addresses**
- Foreign or external IPs not belonging to the organization
- Unusual ports (e.g., non-standard ports like 4444, 1337)
- Repeated connections from the same user/process

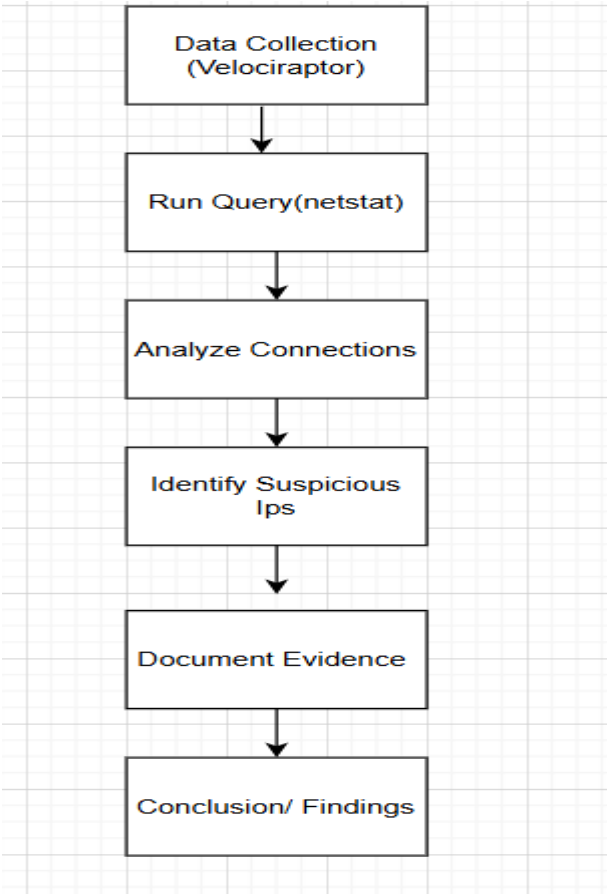
These indicators may suggest:

- Command & Control (C2) communication
- Malware activity
- Unauthorized remote access.

Item ID	Description	Date	IP Address	Port	Hash
E1	Suspicious external Connection detected	2026-04-09	192.168.1.10 -> 45.33.32.156	4444	N/A
E2	Repeated connection attempts	2026-04-09	192.168.1.10 ->78.21.11.90	8080	N/A

Figure 2: Evidence Table

3. <u>Evidence Analysis Flow Diagram:-</u>



4. Conclusion of Evidence Analysis:-

The analysis of network connections revealed suspicious external communications. These connections indicate potential malicious activity such as unauthorized access or command-and-control communication. Further investigation is required to confirm the nature of the threat.